

DLS Buddy — User Manual

Applies to v0.22.0 and later

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1. Introduction

DLS Buddy is a general-purpose Python program for analyzing **dynamic (DLS)** and **static (SLS)** light scattering data. It is not tied to any specific instrument, solvent, or polymer system. All

physical parameters — viscosity, refractive index, temperature, dn/dc , wavelength, and scattering angle — are always supplied by the user. Nothing is assumed or hard-coded.

The graphical interface is organized as a sidebar workspace (your loaded data, auto-grouped into samples) plus tabbed modules:

- **Data** — confirm and edit the physical parameters of each measurement.
- **DLS** — correlation-function fitting (cumulants, exponentials, KWW), CONTIN/NNLS distributions, and multi-angle / concentration-series analysis.
- **SLS** — Zimm, Berry, and Debye analysis for M_w , R_g , and A_2 , with a calibration panel.
- **Cross-Sample** — quantities that link samples or modules: the shape parameter $\rho = R_g/R_h$, and log-log scaling plots.
- **Utilities** — data-quality diagnostics (intensity traces, $I \cdot \sin \theta$), a synthetic correlogram generator, and the Solvent Explorer (§7.2).
- **Settings** — global defaults and appearance.

A core design goal is to support mixed DLS+SLS sessions, linking R_h (from DLS) and R_g (from SLS) measured on the same sample to compute $\rho = R_g/R_h$. The internal data model supports this linkage directly.

Launching the program

Double-click the launcher for your operating system:

- **Windows** — Launch DLS Buddy (Windows).bat
- **macOS** — Launch DLS Buddy (MacOS).command (on first run macOS Gatekeeper may refuse an unsigned script — right-click it → **Open** → confirm once, and it opens normally thereafter)

Or run it from the repository root with:

```
python -m gui.main
```

2. The Workflow at a Glance

Most analyses follow the same short path. Here is the whole flow in one place; the rest of the guide fills in the detail.

Launch → **Load** → **Confirm parameters** + **Update** → **open the DLS or SLS tab** → **pick a method and Run** → **read the results** → **Export**.

1. **Launch** the program (python -m gui.main, or the double-click launcher for your OS).
2. **Load** your data — **Load DLS correlogram...** or **Load SLS intensities...**. Files are auto-detected and auto-grouped into **samples** in the left sidebar (§3).
3. **Confirm the parameters** on the **Data** tab — temperature, viscosity, refractive index, wavelength, scattering angle, and (for SLS) dn/dc and concentration. Identity and optics are entered **once per sample**; press **Enter** to jump to the next field. Click **Update** to commit — analysis runs on the committed values (§6).
4. **Open the analysis tab** for what you need:

- **DLS** (size, R_h): tick the measurement(s) in the sub-tab's list and choose a method — **Correlogram** (cumulant / exponential) or **Distribution** (CONTIN / NNLS / Lognormal) — then **Run**. The multi-angle Γ vs q^2 and concentration-series **D vs c** tabs tick the points to include in the fit (§10.1).
 - **SLS** (M_w , R_g , A_2): select the sample, set the **calibration**, choose **Zimm / Berry / Debye**, then **Run** (§10.2).
5. **Read the results** — each method fills a results **table** plus a plot; flags mark *apparent* vs *thermodynamic* and *uncalibrated* results so they aren't over-read.
 6. **Export** with **Export CSV...** (Origin-compatible) (§11).
 7. (*Optional*) Combine a sample's DLS R_h and SLS R_g on the **Cross-Sample** tab for $\rho = R_g/R_h$ and the scaling plots (§10.3).
-

3. Loading Data

The program accepts data from multiple sources through a **two-layer parser architecture**. Instrument-specific parsers translate native file formats into a common internal structure; all analysis operates on that common structure. Adding a new instrument format never requires changes to analysis or export code.

Every file load follows the same workflow:

1. Click **Load DLS correlogram...** (or **Load SLS intensities...**) and pick the file(s) — you can select **several at once**, and each is auto-detected independently (any that can't be read are reported at the end, the rest still load). The program **auto-detects** the format, trying each supported instrument parser until one recognizes the file.
2. The parser reads the file and pre-fills any parameters it can find.
3. The **Data** tab shows all fields. Change any pre-filled value, and fill in whatever your analysis needs — which fields are required depends on the analysis (e.g. concentration is required for SLS but optional for basic DLS sizing, §6).
4. **Update** commits the parameters; analysis then runs on the committed values.

Note. Pre-filled values from a file are suggestions, not locked inputs. You can always override what the parser found — for example to correct an instrument software value or to apply a calibration you determined independently.

Tip. To set a value on several measurements at once, Ctrl- or Shift-click them in the sidebar, edit the focused one, and press **Update** — your edits apply to every highlighted measurement. (Identity and optics already propagate across a whole sample automatically.)

3.1 Supported File Formats

Loading is **instrument-agnostic**: the load buttons first auto-detect among the instrument formats below (each sniffs its own format). If none match, the program falls back to a **plain two-column** file — for a generic DLS file it asks for the delay-time unit and data form (g_2-1 / g_2 / g_1); a generic SLS file (angle, intensity) needs no prompt. Only if that also fails does it report an unrecognized format.

Format	Parser	Produces
Brookhaven Particle Explorer DLS .CSV	BrookhavenDLSParser	One DLS measurement per file
Brookhaven Particle Explorer count rate .CSV	BrookhavenTraceParser	One intensity trace per file
Brookhaven Particle Explorer SLS .CSV	BrookhavenSLSParser	One measurement per concentration (solvent reference included)
ALV correlator .ASC (e.g. ALV-7012 CGS-12F)	ALVCorrelatorParser / ALVTraceParser	All angles in the file as separate DLS measurements (one sample) + a count-rate trace per angle
Malvern Zetasizer clipboard correlation text	ZetasizerClipboardParser	All records in the file as separate DLS measurements (no metadata — all parameters entered at confirmation)
Malvern Zetasizer structured <i>export</i> .txt/.csv	ZetasizerExportParser	One DLS measurement per row ; pre-fills refractive index, temperature, viscosity, and the sample/material/dispersant names (see §3.2)
Plain-text two-column DLS (delay, $g_2-1/g_2/g_1$)	GenericDLSParser	One DLS measurement per file (fallback; prompts for delay unit + data form)
Plain-text two-column SLS (angle, intensity)	GenericSLSParser	One concentration per file (fallback; loads directly)
Plain-text two-column trace (time, count rate)	GenericTraceParser	One intensity trace per file (Utilities ► Traces; prompts for units)

3.2 Malvern Zetasizer export template

The Zetasizer software’s **export** is configurable, so the program reads its columns by **name** (not by position) and accepts any column order. To produce a file the export parser recognizes, configure the exporter as follows:

- **Settings:** include the header row; use commas as the separator.
- **Parameters (any order):** Sample Name, Material Name, Dispersant Name, Dispersant RI, Temperature, Viscosity, Correlation Delay Times, Correlation Data.

The two correlation blocks (Delay Times + Data) are required; each named parameter is optional and pre-filled when present (the dispersant name is normalized to the controlled solvent vocabulary, §7). The program maps the columns as:

Export column	Pre-fills
Sample Name	sample label

Export column	Pre-fills
Material Name	polymer/material name
Dispersant Name	solvent name
Dispersant RI	solvent refractive index
Temperature	temperature
Viscosity	viscosity
Correlation Delay Times	lag-time axis
Correlation Data	correlogram (g_2-1)

Scattering **angle**, **wavelength**, and **concentration** are not in the export and are entered at the confirmation step, as with any DLS file.

4. Brookhaven Particle Explorer File Formats

Brookhaven Particle Explorer exports DLS and SLS data as comma-separated .CSV files. The formats are described for reference; the parsers handle all reading automatically.

4.1 DLS Correlation Function (.CSV)

- Row 1: sample label string (e.g. `PnVP (40k) in Water - 6/1/2026 4:46:59 PM`).
- Row 2: column headers (`t(μs)`, `C(t)`, `Fitted (CONTIN)`).
- Rows 3+: delay time (μs), $g_2(\tau)-1$, and the instrument CONTIN fit.
- Trailing padding rows with $t = 0$ and $C(t) = -1$ fill unused correlator channels; they are stripped automatically.

Important. The DLS file contains no instrument metadata. Angle, wavelength, refractive index, temperature, viscosity, and sample name must all be supplied manually in the Data tab. **Concentration is optional for DLS** — cumulant, exponential, KWW, and CONTIN/NNLS size analysis (and Γ vs q^2) never use it; it is needed only for the multi-concentration diffusion extrapolation.

Note. The program ignores the Brookhaven CONTIN fit column. All regularization is performed independently by the program's own pipeline.

The correlogram and count-rate history are separate files and are associated manually. (Sample labels in the two files sometimes differ slightly in spacing — a known Brookhaven quirk — which is why automatic pairing is not attempted.)

4.2 Count Rate History (.CSV)

- Row 1: sample label string.
- Row 2: column headers (`Time (seconds)`, `Count Rate (kcps)`).
- Rows 3+: elapsed timestamp (`HH:MM:SS.sssssss`) and count rate in kcps.

Timestamps are elapsed time from the start of the run, not wall-clock time. Count rates are converted to counts per second (cps) on load.

4.3 SLS Intensity Export (.CSV)

One file contains all concentrations and angles, as a flat list of key–value pairs: a metadata header, an angle table, a concentration table, and an intensity grid (**Intensity - Concentration N - Angle M, value**). The parser extracts all available metadata. Only two fields are never present and must be entered manually:

- Sample name.
- Temperature (the SLS export has no temperature field).

Rayleigh ratio. The Brookhaven software embeds its own toluene Rayleigh ratio. It is read and recorded for traceability but **never used** in analysis — the program always computes the temperature- and geometry-corrected value from the toluene Rayleigh literature. See the **Theory-and-Equations-Guide** §1.5.

5. Generic (Plain-Text) File Formats

The generic parsers accept data from any source — home-built correlators, simulation output, collaborator data, spreadsheet exports — as long as it is a clean two-column numerical table. Because these files carry no metadata, **all** physical parameters are supplied in the Data tab.

5.1 Generic DLS Format

File contract (the parser rejects non-compliant files):

- Exactly two columns of numbers; no header, comment, or blank lines.
- Comma- or tab-separated (auto-detected).
- Column 1: delay times (unit specified at confirmation).
- Column 2: correlation values (interpretation specified at confirmation).
- At least two data rows.

Delay-time units: s, ms, us (μ s), ns — converted to seconds on load.

Column-2 interpretation (data form). The program converts every input to $g_2(\tau)-1$ (baseline-subtracted intensity autocorrelation) before storage:

Data form	Column 2 contains	Extra parameter
g2m1	$g_2(\tau)-1$, already baseline-subtracted	none
g2	$g_2(\tau)$, baseline not yet subtracted	baseline $B > 0$
g1	field autocorrelation $g_1(\tau)$	coherence factor $\beta \in (0, 1]$

For the g2 form, the baseline is removed; for the g1 form, the Siegert relation converts the field autocorrelation to the measured quantity (β is the instrument coherence factor, an optical property, typically 0.01–1.0 for standard goniometers, determined independently). See the **Theory-and-Equations-Guide** for the equations.

Tip. Unsure which form you have? Check the range of column 2. $g_2(\tau)-1$ starts near the intercept β (often 0.3–1.0) and decays to 0; unsubtracted $g_2(\tau)$ starts near $1+\beta$ and decays to 1; $g_1(\tau)$ starts near 1 and decays to 0.

5.2 Generic SLS Format

Each file is one concentration at multiple angles (one file per concentration).

- Exactly two columns; no header/comment/blank lines; comma- or tab-separated.
- Column 1: scattering angles in degrees, each strictly in (0, 180).
- Column 2: raw intensity values.
- At least two data rows.

Column order. Angle must be column 1, intensity column 2. If reversed, the parser detects column-1 values outside (0, 180)° and raises an error suggesting the columns may be swapped.

For a Zimm/Berry/Debye dataset, load one file per concentration (including the solvent reference at $c = 0$) and set each concentration in the Data tab. The SLS module assembles the dataset from the loaded measurements.

6. Physical Parameters and Units

All quantities are **stored in canonical internal units**; you enter and read them in whatever unit is convenient. In the **Data** tab each parameter row has a **Unit** column: a dropdown for quantities with a choice of units (the value re-displays when you switch units — the stored physical value is unchanged), or a fixed label for quantities with a single natural unit.

The identity and label fields carry no units; the physical quantities do. The **Purpose** column notes what each field feeds:

Parameter	Internal unit	Input units (default first)	Purpose
Label	— (text)	—	Free-text tag for your own reference; does not affect grouping or analysis
Solute name	— (text)	—	Names the scatterer (need not be a polymer); part of sample identity
Solvent name	— (text)	—	Part of sample identity; picks library n/η for common solvents (§7.1)
Mw fraction	— (text)	—	Optional label separating molecular-weight fractions of one polymer (§8.1)

Parameter	Internal unit	Input units (default first)	Purpose
Polarization (DDLS)	— (VV/VH/VU)	—	Tags a correlogram’s polarization for depolarized analysis (§10.1); blank = ordinary DLS
Concentration	g/mL	mg/mL , g/L, µg/mL, g/mL	Required for SLS and the DLS concentration series
Temperature	kelvin (K)	°C, K, °F	Sets Stokes–Einstein Rh and library properties
Viscosity	Pa·s	mPa·s , cP, Pa·s	Solvent viscosity η for Stokes–Einstein
Solvent refractive index	— (dimensionless)	—	Sets the scattering vector q ; library-proposed for common solvents (§7.1)
Wavelength	nanometers (nm)	nm	Laser wavelength; sets q
Scattering angle	degrees	deg	Detector angle; sets q
dn/dc	mL/g	mL/g	Refractive-index increment (SLS); the most sensitive SLS input
Delay-time window (DLS tab)	seconds (s)	µs , ns, ms, s	Restricts which lags are fitted
Count rate (trace)	counts/second (cps)	kcps , cps, Mcps (chosen on load)	Intensity-trace units

SLS intensities. The calibrant intensity and dark count rate are entered in **the same units as your SLS intensity file** (whatever the instrument exported). The program does not convert them, because only the *ratio* of calibrant to sample intensities enters the Rayleigh ratio — a fixed-unit conversion there would introduce a scale error. The fields are labeled “(file units)” accordingly.

7. Solvent Names

Solvent identity is part of sample identity: the same polymer at the same concentration and temperature in two solvents is two distinct datasets (enabling the comparison of the same polymer in different solvents). Names are normalized against a controlled vocabulary, so **Water**, **water**, and **H2O** are equivalent. Unrecognized names are accepted (used lowercased) with a one-time warning; to add one permanently, edit `_SOLVENT_ALIASES` in `core/data_models.py`.

Note. Sample-name matching is case-sensitive (PVP $\neq$ pvp); solvent-name matching is case-insensitive and alias-aware. (The sample name is stored internally as the `polymer_name` field — samples need not be polymers.)

7.1 Automatic refractive index and viscosity

For common solvents the program can **propose** the two optical/hydrodynamic properties every scattering calculation needs — the **refractive index** n (which sets the scattering vector) and, for DLS, the **viscosity** η (which sets the Stokes–Einstein radius) — so you need not look them up by hand. In the **Data** tab the *Solvent name* row is a dropdown; pick one of the library solvents and n (and η) fill in automatically from the sample’s temperature and wavelength, each marked with a small **teal dot**.

- The values **re-derive** whenever you change the temperature or wavelength.
- Each proposed value is **rounded at the last digit the library can stand behind**. The uncertainty itself only chooses that place; it is never used in any analysis.
- A value you **type by hand is never overwritten** by the library, and shows no dot (to return to the library value, clear your entry and re-pick the solvent).
- Outside a solvent’s validity range the program **does not guess**: it clears the library value and shows a note asking you to enter it manually.

Only the library’s higher-confidence (**primary**) solvents auto-fill into a measurement; a free-text solvent name leaves the fields for you to complete. The primary solvents and their validity ranges are tabled in §7.3; the full list, with the uncertainties and sources, is in the **Theory-and-Equations-Guide** (§5).

7.2 The Solvent Explorer (Utilities ► Solvent Explorer)

The **Solvent Explorer** — a sub-tab of the **Utilities** tab — is a standalone look-up and visual calculator over the whole library. It is global — it needs no loaded sample, ignores the sidebar selection, and never writes into a measurement. Choose a solvent, temperature, and wavelength to read n and η with their uncertainty and a **confidence-tier badge** (all library solvents appear here; lower-confidence ones are labeled *estimate*). Two plots, side by side on one shared figure, show at a glance the range of conditions the library can serve:

- η and n vs temperature (twin axes — viscosity on the left, on its own log-capable scale; refractive index on the right), and
- n vs wavelength,

each drawn **only across the range the library covers**, with a shaded **confidence band** and a marker at your chosen condition. The band is a conservative bound **at each condition** — narrowest near the conditions the sources measure best, growing toward the edges of the validity range, and honestly flat where a source states no temperature shape (see the **Theory-and-Equations-Guide** §5). Like every library uncertainty it is **displayed, never folded into a reported $\pm$** . Set which solvent the Explorer opens on under **Settings** $\rightarrow$ **Solvent library** $\rightarrow$ **Default solvent**.

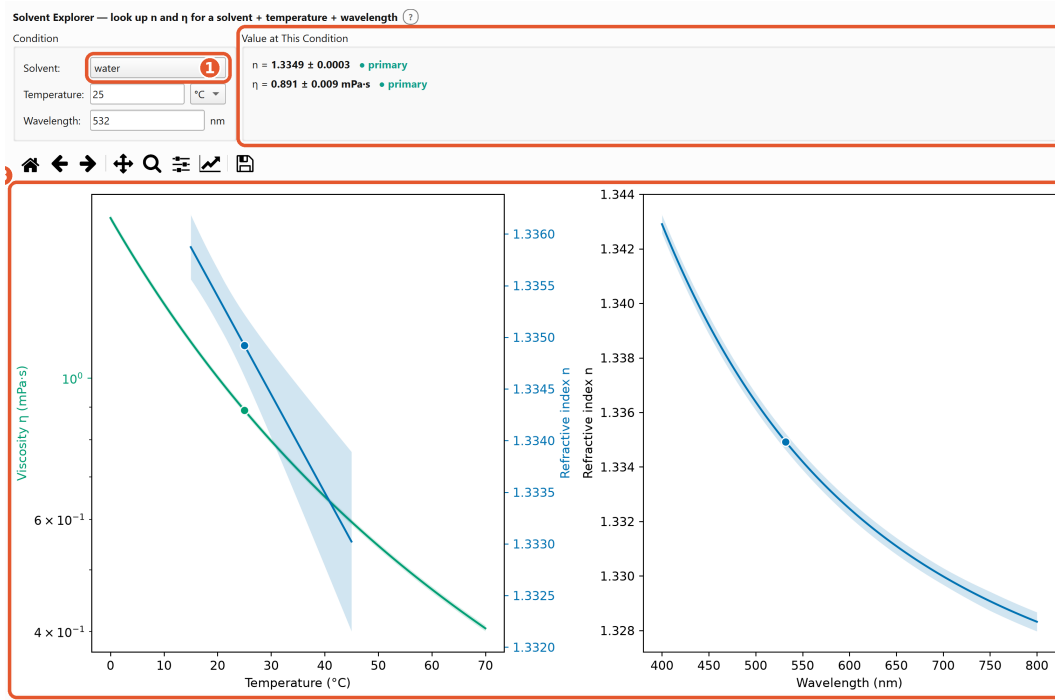


Figure 1: The Solvent Explorer (Utilities ► Solvent Explorer): pick a solvent (1); read n and η at the chosen temperature and wavelength with their uncertainty and a confidence-tier badge (2). The plots (3) draw η and n vs temperature and n vs wavelength, each with a shaded confidence band, only across the range the library covers.

7.3 The Solvent Library at a Glance

The table below lists the **primary**-tier solvents — the higher-confidence set that auto-fills n and η in the Data tab (§7.1) — with the range of conditions each one covers. Ranges are **per-solvent**, not uniform: they follow the span of the underlying measurements, so refractive index and viscosity may be valid over different temperature windows for the same solvent. Outside a listed range the program stops proposing a value and asks you to enter it by hand (§7.1). The uncertainties behind these ranges, and the sources they come from, are in the **Theory-and-Equations-Guide** §5.

Solvent	RI λ -range (nm)	RI T -range (°C)	η T -range (°C)
acetone	476–633	15–40	–18–46
benzene	476–633	15–40	15–66
cyclohexane	476–633	15–40	15–76
ethanol	476–633	15–40	–20–70
ethylene glycol	400–800	1–45	0–71
glycerol	400–800	1–45	15–51
hexane	400–640	15–40	–3–66
methanol	450–700	15–50	0–70
toluene	405–830	15–40	–13–96
water	400–800	15–45	0–70

The library also carries 9 **lower-confidence** (*estimate-tier*) solvents, drawn from thinner or less consistent source data: acetonitrile, ccl4, chloroform, cs2, dcm, dmf, dmso, ethyl acetate, and thf. They appear in the Solvent Explorer (labeled *estimate*) but are held back from the Data-tab auto-fill, so a value only ever reaches an analysis when you pick it there yourself.

8. How the Program Organizes Data

Every measurement carries a sample key — (polymer, solvent, concentration, temperature) — used to associate related datasets, for example to pair a DLS and an SLS measurement of the same sample when computing $\rho = Rg/Rh$. For grouping into a **sample**, the identity is (polymer, solvent, rounded temperature); concentration and angle are axes *within* a sample.

The **Workspace list** on the left is your navigator and data manager: click a measurement to load it into the active tab, click a sample or DLS/SLS heading to select everything under it, and right-click to remove, move/regroup, or average replicates. It only *navigates* — each analysis tab decides what it actually analyzes (you **tick** measurements, or pick a sample from the tab's own **Sample** dropdown). The list also **mirrors** the active tab's current selection — those measurements read **bold blue** — and flags a **not-yet-confirmed sample in amber** (set its identity in the Data tab and press *Update*).

8.1 Molecular-weight fractions

A scaling study measures several **molecular-weight fractions** of one polymer (same chemistry, solvent, and temperature), each with its own concentration series. Because molecular weight is *not* part of the sample identity, all such fractions group into one sample. To keep them separate, give each measurement an optional free-text *Mw* fraction label (e.g. 250k, 1M) in the Data tab; editing it offers to apply the label to every concentration loaded from the same file. The label is *not* required — leave it blank for single-fraction samples or systems with no meaningful *Mw* (e.g. nanoparticles).

Within a sample, the fraction label **partitions the SLS analysis**: the SLS tab shows a *Fraction* selector and runs an independent Zimm/Berry/Debye fit for the selected fraction, while the single $c = 0$ solvent reference is **shared** across all fractions. Each fraction therefore yields its own *Mw*, *Rg*, A_2 (and, from DLS, *Rh*). In the Cross-Sample tab each fraction contributes its own point to the scaling plots (*Rg-Mw*, A_2 -*Mw*) and its own row in the ρ table, so a molecular-weight series produces a proper scaling curve rather than a single merged point. (The per-fraction Cross-Sample views build on the per-fraction SLS analysis described above.)

Intensity traces (count-rate histories) are independent objects: they can be loaded and analyzed without a correlogram, or linked to a DLS measurement after the fact — supporting batch diagnostics over a concentration series.

9. Data Quality and Utilities

The Utilities module assesses data quality (especially for intensity traces) and generates synthetic test data. These help identify slow-mode fluctuations, drift, and dust before committing to full correlation-function or Zimm analysis.

9.1 Intensity-Trace Diagnostics

- **Summary statistics & baseline.** Mean, standard deviation, coefficient of variation, min, max. The baseline (true scattering level) is estimated robustly so upward dust spikes do not bias it — by a lower percentile (default 25th) or a sigma-clipped mean.
- **Outlier flagging.** Points outside the shot-noise envelope are flagged. For a Poisson (shot-noise-limited) process the standard deviation equals $\sqrt{\text{mean}}$, so this is the expected $\pm k\sigma$ band (default $k = 1$). The flagged fraction is a compact quality metric.

Interpretation. $\sqrt{\text{mean}}$ equals the true shot-noise σ only for raw photon counts; for count rates (cps) the envelope is a useful heuristic.

- **Running average.** Sliding-window mean and standard deviation (time- or point-based window). The time-based window handles irregular or non-uniform sampling; a wide window (30–60 s) separates slow drift from shot noise.
- **Block-variance (blocking) analysis.** The standard error of the mean vs. block size: flat for an uncorrelated trace; a rise-then-plateau is the signature of positive correlations (slow modes). The full curve is returned with a correlation flag (robust plateau-vs-small-block ratio, default threshold 1.5 — a general heuristic, not a system-specific cutoff).
- **Count-rate histogram.** Fitted with a Gaussian and/or Poisson distribution (Freedman–Diaconis binning, Sturges fallback). The key diagnostic is the **Fano factor** (variance/mean in count space): ≈ 1 is shot-noise limited (ideal Poisson); > 1 signals excess fluctuations (slow modes, dust, drift). Reduced χ^2 of each fit and the coefficient of variation (vs. the shot-noise expectation $1/\sqrt{N}$) are also reported. See the **Theory-and-Equations-Guide** for the equation.
- **Stationarity test.** An augmented Dickey–Fuller test gives a p -value for whether the trace is stationary; non-stationarity indicates drift or slow-mode behavior. Requires `statsmodels`.

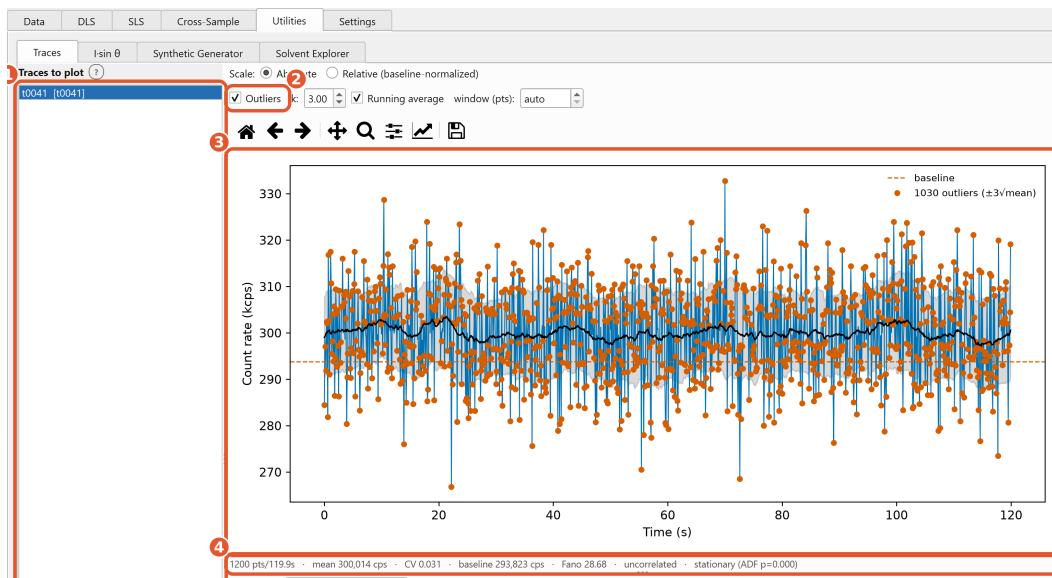


Figure 2: Utilities ► Traces. Select the trace(s) to plot (1) and toggle the overlays (2). The count-rate-vs-time plot (3) shows the robust baseline (dashed), flagged outliers (dots), the running average (solid), and the shot-noise band (shaded); the summary line (4) reports the point count, mean, coefficient of variation, baseline, and the Fano factor.

9.2 SLS $I\text{-sin}(\theta)$ Diagnostic

For an ideal, isotropic, dust-free scattering volume, $I\text{-sin}(\theta)$ is flat across angle. Curvature, asymmetry about 90° , or upturns at the extremes reveal alignment problems, stray light, or dust. Curves can be shown **absolute** (comparing concentrations) or **normalized** (each divided by its mean — for comparing solvents, e.g. toluene vs. water, where absolute scale is meaningless). Choose which sample to plot from the **Sample** dropdown at the top of the tab (only samples with angular-intensity data — SLS or a solvent reference — appear).

Tip. A flat $I\text{-sin}(\theta)$ is much harder to achieve with water than toluene. Overlaying normalized curves for both is an effective alignment/stray-light check.

9.3 Synthetic Data Generator

Generates artificial light-scattering data with **ground-truth** parameters — to validate the analysis (recover a known answer) or produce test files. The forward model uses the **same** q , Stokes–Einstein, optical-constant, Rayleigh, and Zimm physics as the analysis code, so loading and analyzing the output recovers what you put in (no convention mismatch).

Inputs. *DLS* and *SLS* need different physical parameters, so they have separate input blocks that share the optical **conditions** (wavelength, n , temperature, viscosity) and the angle/concentration grid:

- **DLS sizes** — a populations table (median R_h , relative scattering weight, log-normal spread; one row = monomodal, two = bimodal, ...) plus the coherence factor β , noise level, and point count. The populations *are* the modality — there is no separate “bimodal” option, and the CONTIN/NNLS distribution is something you then recover by analyzing the generated correlogram, not a generated artifact.

- **SLS sample** — M_w , R_g , A_2 , and dn/dc .
- **Calibration** — *Default* writes calibration metadata into a saved SLS file (and, on workspace injection, prefills the session calibration so M_w is absolute); *Uncalibrated* omits it, so an analysis run is flagged arbitrary-scale.

Artifacts. Tick **Preview** and/or **Save** per artifact:

Artifact	What it tests	Saved as
Correlogram (one angle)	cumulant / single / double / KWW / CONTIN / NNLS	generic two-column CSV
Count-rate trace	trace diagnostics (outliers, drift, Fano, ADF)	two-column CSV
Multi-angle DLS (all angles)	Γ -vs- q^2	multi-angle .ASC (+ a trace per angle)
SLS — full Zimm set	Zimm / Berry (thermodynamic)	multi-concentration SLS .csv
SLS — single concentration (angular)	Debye / Guinier (apparent)	SLS .csv
SLS — single angle (conc. series)	calibration-free A_2 / single-angle M_w	SLS .csv
Full M_w -series test set	everything (a homologous series)	a Clean + a Messy folder

Preview draws the *raw generated data* (the analysis figures come from loading the data into the analysis tabs). **Save** writes loadable instrument files into the chosen folder. **Add to workspace as sample** injects a coherent sample (the primary DLS artifact, the primary SLS artifact, and any trace) under a parsed **name** / **solvent** identity, so DLS + SLS connect for $\rho = R_g/R_h$ without touching disk. Preview never changes any state.

10. Running an Analysis

This section summarizes, procedurally, how to use the three analysis tabs. For the equations, derivations, and literature behind each method, see the **Theory-and-Equations-Guide**.

10.1 DLS tab

The DLS module extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$. The tab is organized into **sub-tabs**, each a persistent full-size view (running one never clears another):

- **Correlogram** — parametric fits (cumulant / single / double / KWW). The correlogram is shown in **four scales at once** — a large lin-log view plus lin-lin / log-lin / log-log thumbnails; **double-click a thumbnail to promote it** to the main view. A residual panel sits below, and **Cumulant** and **KWW** fits each populate their own results table. The raw correlogram appears as soon as you tick a measurement, before any fit. The **cumulant order** is set here (seeded from Settings; the per-run value wins). The **cumulant method** is a global choice in **Settings** → **Cumulant method**: *Nonlinear* (Frissen, the default — fits g_2-1

directly with a floating baseline, more robust to drift and noisy data) or *Linear* (Koppel — the classic log-polynomial fit). It applies to every cumulant-based step (cumulant, Γ vs q^2 , replicate averaging); the table shows which method produced each result (and its fitted baseline). Switching the method clears existing cumulant-based results — you are asked to confirm first, and distributions, SLS, and any hand-entered R_h are kept. A **delay window** (min/max, in μ s) restricts which lags are fitted. Separately, **Settings** $\rightarrow$ **Skip initial lag channels** drops the first N channels (the shortest lags) from *every* DLS fit — set it to your correlator’s afterpulsing count when a fit is dragged to a falsely small size by a noisy first channel (default 0; the skip and the delay-window minimum compose, the later of the two winning).

- **Distribution** — NNLS / CONTIN / Lognormal (tick one or more **methods** to compute, with the R_h/Γ axis toggle). These recover a full distribution of decay rates without assuming a functional form; NNLS resolves well-separated peaks but is spiky, CONTIN is smoother and generally better for broad/real data, and Lognormal is a robust single-mode default for a known-unimodal sample. Resolved **peaks are labeled** on the plot, and a **residuals panel** (data — reconstructed g_2-1 vs delay time) sits below it. The R_h grid (min / max / points) and the **CONTIN L-curve** α range are set here (seeded from Settings; the per-run value wins). The sub-tab has its own **Peak results** table (rows *1st peak* / *2nd peak* / ..., one color-matched column per ticked measurement·method, cells reading e.g. “12.3 nm · 87%” — the peak R_h and its intensity-weighted area), mirroring the Correlogram results panel; the same values also roll up into the workspace-wide **Summary** sub-tab.
- **Γ vs q^2 and D vs c** — sample-level analyses. “ Γ vs q^2 (sample, multi-angle)” tests whether motion is genuinely diffusive and returns D (and R_h); “D vs c (sample, multi-conc.)” extrapolates the apparent diffusion coefficient to zero concentration, returning the infinite-dilution $R_{h,0}$ and the interaction parameter kD . Both operate on the selected sample’s DLS measurements (of the current M_w fraction). Each tab carries a “**Measurements (tick to include in the fit)**” points table — one row per measurement (Angle · c · Γ · D_{app} for Γ vs q^2 ; c · Angle · D_{app} for D vs c) — and the **tick boxes choose which points enter the fit**. On focus the table lists the sample’s measurements as metadata only (Γ and D read “—”), and it **refreshes as soon as you commit** in the Data tab — no sidebar re-click needed. Every eligible row is ticked by default; changing the selection marks a shown result “*selection changed — press Run to refresh*” rather than refitting live or clearing the plot. Bulk helpers speed selection: **Select all**, **Select none**, and **Tick all at** a chosen value — a dropdown of the distinct **concentrations (in mg/mL)** on Γ vs q^2 and **angles (in °)** on D vs c. (The **Distribution** sub-tab has both bulk selectors too.) Rows that can’t be ticked are grayed with a **hover tooltip explaining why** (parameters not confirmed, or a different M_w fraction); a note also flags any same-polymer/solvent measurements held in a separate sample because their temperature isn’t confirmed yet. **Run** stays disabled until at least two distinct angles / concentrations are ticked (a status line reads “*Tick at least two distinct angles to run.*”). After a Run the Γ/D_{app} columns fill in, a point whose cumulant fit failed or whose PDI exceeds the validity limit is flagged in its tooltip, and unticked points gray out on the plot. Scalar outputs fill a **Results** table (D , R_h , R^2 , *Diffusive?* for Γ vs q^2 ; D_0 , $R_{h,0}$, kD , R^2 for D vs c). The Γ and D used here come from an **internal 2nd-order cumulant fit** of each correlogram — independent of any saved Correlogram/Distribution result — using the global skip-channels and cumulant method from Settings.
- **Depolarized (DDLs)** — pairs a VV with a VH correlogram at each angle (tag each correlogram).

ogram’s polarization in the Data tab) to separate translational from rotational diffusion, and fits sphere and rigid-rod shape models for particle dimensions. A **Paired angles** checklist ticks which paired angles enter the Dr/Dt fit (all ticked by default; untick an outlier), and *Include all angles* re-ticks them. Run with *Run DDLS*; the sphere model needs the solvent viscosity. The rotational-diffusion results and the sphere/rod shape-model outputs each fill their own **results table**.

Across all of the DLS analysis tabs above: if you go back to the Data tab and **commit a change to a parameter that fed a fit already on screen**, the plot stays but a status line reads “Inputs changed since this fit — press Run to refresh” — so a result is never silently shown against parameters it wasn’t computed with. The hint is specific to the results whose own inputs changed. (The **SLS tab behaves the same way** — switching sample/fraction or committing redisplay the last fit and flags it stale; only hiding/showing a masked point refits on its own. See §10.2.)

- **Summary** — a workspace-wide, exportable results table that **persists across save/reload**. A *per-measurement* table (one row per measurement) shows cumulant $Rh + PDI$, single/KWW Rh , and NNLS/CONTIN peaks as Rh (Int %) (Int % is the intensity-weighted peak area, *not* a mass/weight percent). A sample-level Rh table collects replicate averages, $\Gamma-q^2$ ($q \rightarrow 0$, **apparent**) and $D-c$ ($c \rightarrow 0$, **thermodynamic**) — the Rh Type column marks that distinction and **From** lists the contributing measurements. The left checklist is shared with the Correlogram/Distribution tabs and doubles as a “**Ticked only**” filter; **Export CSV...** writes one long/tidy row per result or peak. Rows fill in as you run fits and averages (e.g. *Average derived results* from the sidebar).

Co-plotting several measurements. Both the Correlogram and Distribution tabs carry a “**Measurements to plot**” checklist of every loaded DLS measurement (any sample, angle, concentration, or Mw fraction); tick several to overlay them for comparison. This checklist — **shared between the two tabs** and mirrored in the Workspace list (ticked measurements read bold blue) — is the **sole driver** of what gets fit; selecting a measurement in the Workspace list only *navigates*. **Run** fits every ticked measurement (and, on the Distribution tab, every ticked **method**). Fitting many heavy distributions (especially CONTIN) at once is slow, so tick only what you want to compare.

Every single-axes DLS/SLS analysis plot carries an **axis-control bar** beneath it: per-axis linear/log toggles, explicit min/max boxes, and an autoscale button.

Delay-time window + baseline region. A single shared analysis region (set on the **Correlogram** sub-tab and used by all per-measurement DLS methods, on every ticked measurement) defines the fit window and the baseline region. Set them by typing values or by **dragging the markers on the correlogram**: the green dotted lines are the window (points outside are ignored — useful to trim a noisy long-delay tail), and the gray dashed/shaded span is the **baseline region** used by the distribution methods (parametric fits don’t use a baseline). When a window and a baseline marker overlap, tell them apart by their **offset caret handles**: the **window handles are green carets along the top** of the plot, the **baseline handles are gray carets along the bottom** — grab the top or bottom half of the plot to pick the one you want. Defaults: full range (the union of all ticked measurements), and the last 25 % of delays for the baseline. A ticked measurement whose lag axis doesn’t overlap the window (e.g. a millisecond-lag set co-plotted with a microsecond-lag one) can have too few points inside — it is then

skipped with a note; drag the markers to a suitable range.

Procedure. Tick one or more measurements in the checklist, set the delay window and baseline, choose a method (or tick the distribution methods you want), click **Run**, and read the result in the table and plot. See the **Theory-and-Equations-Guide** for the equations and theory.

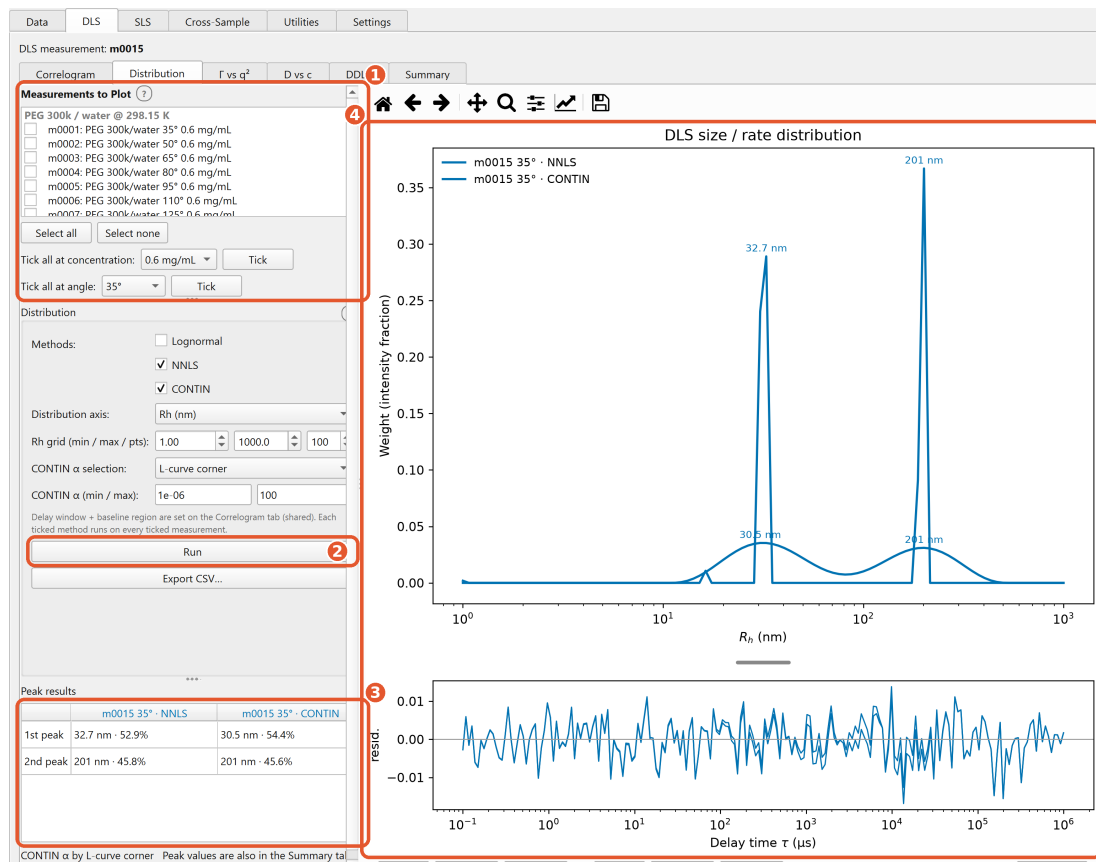


Figure 3: The DLS **Distribution** sub-tab on a bimodal sample. Tick the measurement(s) and the methods (1) and press **Run** (2); each method's size distribution is overlaid on the plot (4) with its peaks labeled, the residuals below, and the resolved R_h and intensity fraction of every peak in the **Peak results** table (3). NNLS gives sharp peaks, CONTIN a smoother curve.

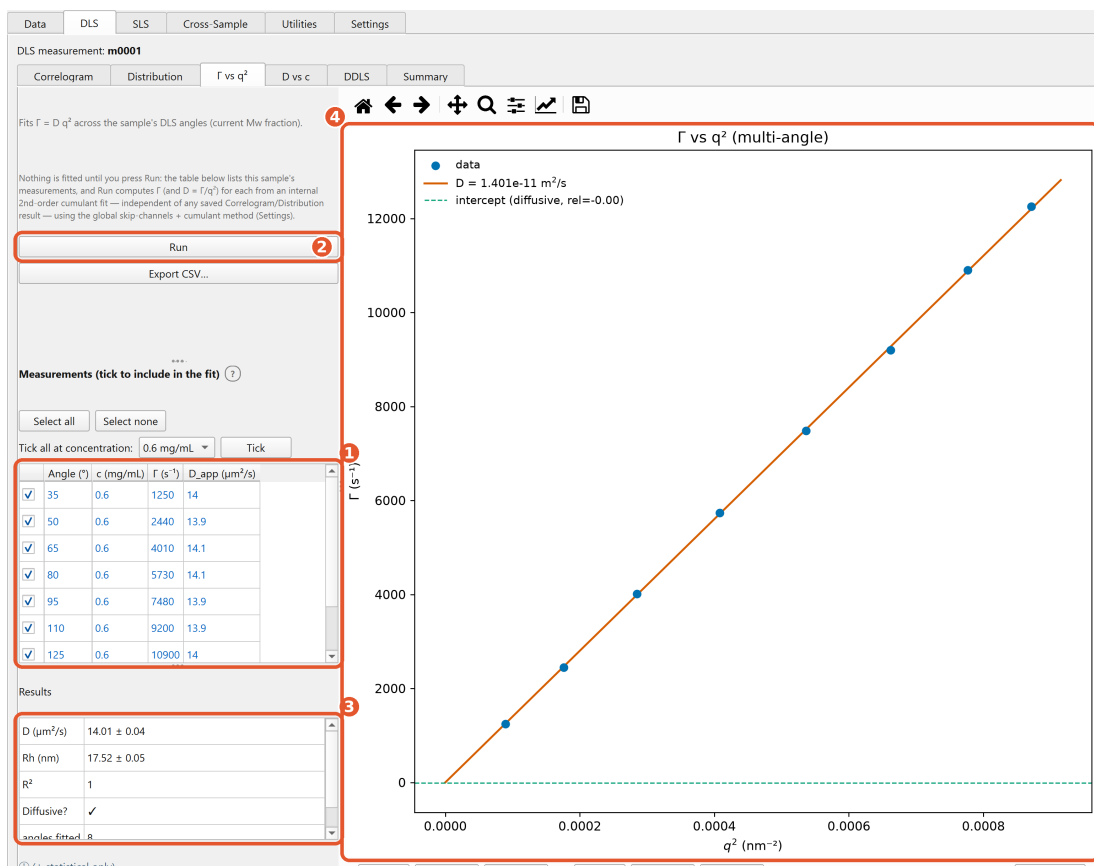


Figure 4: The DLS Γ vs q^2 sub-tab. Tick which angles enter the fit (1) and press **Run** (2); the decay rate Γ is fitted against q^2 across the sample's angles (4). A straight line through the origin confirms diffusive motion; the results table (3) reports D , R_h , R^2 , and the diffusive check.

10.2 SLS tab

The SLS module extracts the weight-average molecular weight M_w , the radius of gyration R_g , and the second virial coefficient A_2 . It first converts raw intensities to absolute excess Rayleigh ratios (via the calibration panel), then constructs the Zimm, Berry, or Debye plots. Pick which sample to analyze from the **SLS sample** dropdown at the top of the tab (only samples with SLS data appear); the whole Zimm set — solvent reference plus the concentration series — is analyzed together.

Calibration panel. The program computes its own calibration constant kc from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard's geometry-aware Rayleigh factor. The constant is shown and can be edited. Any value stored in the instrument file is kept only for reference and **never** enters the calculation. If no kc can be supplied, the analysis still runs on an arbitrary scale, flagged uncalibrated — then M_w and the absolute A_2 are marked unreliable, but R_g and the calibration-free product $2A_2M_w$ remain valid.

Methods available.

- **Single-concentration (apparent) Debye** — at one concentration, gives an apparent M_w and apparent R_g . A single angle yields only an apparent M_w and no R_g . Apparent values

are never presented as final.

- **Zimm / Berry double extrapolation** — combines several concentrations and angles and extrapolates to both zero angle and zero concentration, giving Mw , Rg , and A_2 at once. The result is marked **thermodynamic**. Berry linearizes the high- qRg regime (large Mw) better than Zimm; the program reports which construction it used.
- **Calibration-free A_2** — obtains the product $2A_2Mw$ from ratios of excess intensities across concentration at a fixed angle, without any absolute calibration. Valuable when absolute calibration is uncertain.
- **Guinier** — a single-concentration route to Rg from the log intensity, scale-independent and reliable even on an uncalibrated run.
- **Rayleigh series** — the excess Rayleigh ratios themselves.

Spacing constant (Zimm/Berry grid). Set the purely **aesthetic** spacing constant in the **Spacing k** box (or press **Suggest** for the automatic value). It only changes how spread-out the curves look — Mw , Rg , and A_2 come from the real fit and are unaffected.

Data masking. You can hide whole angles/concentrations or individual points (click to mask, grayed and persisted); an over-masked Zimm is redirected to Debye/Guinier. A manual- Mw override is available — useful when the trustworthy Mw was characterized independently (say, in a different solvent or by GPC/MALS) and you want the analysis to use that value — and it is never overwritten by a re-analysis.

Depolarization correction. For optically anisotropic scatterers, measuring the depolarized (VH) component enables a Cabannes correction that keeps Mw from being overestimated, and yields the optical anisotropy as a result in its own right. Enter IVH and press **Compute depolarization**; the correction is **display-only** and is never written back to the sample's Mw .

Results. Each method fills a two-column **Results** table — Zimm/Berry give Mw , Rg , A_2 and R^2 ; Debye/Guinier the apparent values; single-angle an apparent Mw ; calibration-free A_2 the $2A_2Mw$ product. The depolarization correction has its **own table** (ρ_v , ρ_u , δ^2 , Cabannes f , anisotropic %). The Rayleigh series has no scalar table — it shows a one-line status (“*Excess Rayleigh ratio for N concentrations.*”) alongside its plot. **Run analysis is the only thing that computes a fit** (the same explicit-Run model as the DLS tabs): switching sample, fraction, or method **keeps the last result on screen** rather than blanking it, and if you commit a parameter that fed the shown fit a status line flags it “*Inputs changed since this fit — press Run to refresh*” instead of silently recomputing. The one exception is **masking** — hiding or showing a point re-runs the fit immediately.

Procedure. Set the calibration in the panel (or accept the uncalibrated fallback), choose the fraction (if any), select a method, mask any bad points/angles, click **Run analysis**, and read Mw / Rg / A_2 with their apparent/thermodynamic and calibrated/uncalibrated flags. See the Theory-and-Equations-Guide for the equations and theory.

10.3 Cross-Sample tab

The Cross-Sample module computes quantities that link samples or modules. It is *aggregate-scoped*, so it makes **two** selections, both in the tab:

- **Membership** — an **include/exclude** list (left) of every sample with SLS data (all start included); untick to drop a sample from the ρ table and scaling plots.

- **Focus** — the source panel’s own **Sample** and **Fraction** selectors, which say exactly which sample and Mw fraction you are editing. This is explicit: the shell Workspace tree only *highlights* the focused sample (read-only) — clicking it there does not drive this tab, and the ρ table is a read-only results view.

$\rho = Rg/Rh$. The shape parameter links Rg (from SLS) and Rh (from DLS) of the same sample. Each included sample (or *fraction*) gets one row: Rg , Rh , ρ , a **Shape** label naming the likely architecture, and whether ρ is **apparent** or **thermodynamic**. The program checks that the two measurements share polymer, solvent, and temperature.

Sample	Rg (nm)	Rh (nm)	$\rho = Rg/Rh$	Shape	Type
1 PEG 300k / water @ 298.15 K	26.03	17.54	1.484 ± 0.011	random (Gaussian) coil (~1.5)	apparent

Figure 5: The Cross-Sample $\rho = Rg/Rh$ table. Tick which samples to include (1); each included sample gets one row (2) with its Rg , Rh , ρ , a Shape label, and an apparent/thermodynamic flag. A per-row dropdown (not shown) chooses the source for each quantity.

Because a sample can have several Rg and Rh values, you **choose the source** for each (a dropdown per row), and the program never picks silently:

Quantity	Sources (best-tier → apparent)	Hand-entry?
Rg (SLS)	Zimm/Berry double extrapolation (<i>thermodynamic</i>); single-concentration	Yes
Rh (DLS)	Debye/Guinier (<i>apparent</i>) $c \rightarrow 0$ D -vs- c extrapolation (<i>thermodynamic</i>); $q \rightarrow 0$ Γ -vs- q^2 at one concentration (<i>apparent</i>); replicate average (<i>apparent</i> — the one dynamic Rh carrying a real $\pm$ SE, so preferred over a single cumulant and never overwritten by a refresh); single-condition cumulant (<i>apparent</i>); CONTIN/NNLS distribution peak (<i>apparent</i>)	Yes
Mw (SLS)	Zimm/Berry extrapolation; single-concentration Debye/Guinier — calibration-dependent (an uncalibrated fit is on an arbitrary scale, flagged below)	Yes

Quantity	Sources (best-tier → apparent)	Hand-entry?
A_2 (SLS)	Zimm/Berry double extrapolation only (no single-concentration A_2) — calibration-dependent	No (picker only)

A_2 has **no hand-entry** because it is a solvent/temperature-specific interaction coefficient with no external “standard”, and it is the very quantity the A_2 – Mw plot fits. A hand-entered Rg , Rh , or Mw is never overwritten by a later re-analysis.

Each picker groups its candidates by result type (extrapolated / distribution / averaged / single-condition) and hides the long tail of per-angle / per-concentration *apparent* values behind a “**Show all single-condition results**” checkbox. The best-tier value stays selected either way.

Calibration flag. Mw and A_2 are on an arbitrary scale until the run is calibrated, so each row shows an “**uncalibrated**” warning badge and the panel a one-line banner when the chosen value is uncalibrated. Rg is scale-independent and is never flagged. For an *SLS-only* sample the Rh (DLS) row is shown but disabled.

Picking a peak. If you have run a CONTIN or NNLS distribution for a measurement in the DLS tab, each resolved population appears as its own Rh source in the dropdown, so you choose which population feeds ρ . The dominant peak can be the default; a minor peak is selectable but never chosen automatically. The percentages are intensity fractions, not number or mass fractions. Distribution peaks are surfaced only from a distribution you have already computed.

The **default** source is the most-extrapolated the data support (a labeled choice), with fit quality as the tiebreak. ρ is **flagged apparent if either input is apparent**. For a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes.

Units. The Rg/Rh columns and the manual-entry fields follow the global **Settings** → **Display units** choice (e.g. nm or μm for size, g/mol or kg/mol for Mw); the scaling-plot axes use the same setting. Everything is stored in canonical units and only converted for display, and a value you type is read in the shown unit.

Scaling laws (Rg – Mw , A_2 – Mw). The **Scaling** sub-tab shows two stacked plots over the included samples (one point per sample/fraction), using each sample’s effective Mw (a hand-entered Mw wins) chosen in the same source panel. The lower plot is always A_2 – Mw ; the upper plot is a **size** scaling whose axis you swap between Rg – Mw (SLS) and Rh – Mw (DLS) with the **Show Rh – Mw / Show Rg – Mw** button. Below each plot the tab prints a plain-language reading of the fitted exponent. Two cautions the tab surfaces: an **uncalibrated** Mw is on an arbitrary scale (its horizontal position is meaningless — enter a trusted Mw instead), and A_2 can be **negative** (below the θ -point), in which case that point cannot sit on the log A_2 axis and is excluded. Curating which samples form the series is the user’s responsibility — mixing polymers or solvents on one fit is not meaningful.

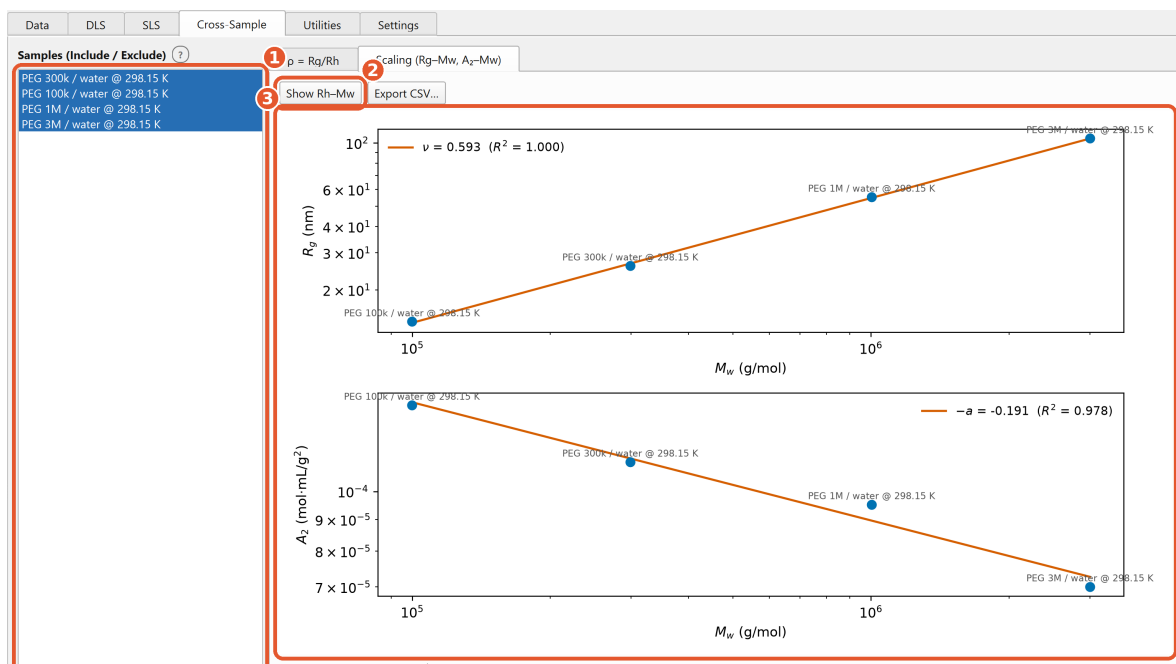


Figure 6: The Cross-Sample **Scaling** sub-tab over a PEG molecular-weight series. Each included sample (1) is one point; the upper plot fits R_g – M_w (swap to R_h – M_w with the button (2)) and the lower plot A_2 – M_w , both on log–log axes (3). The fitted exponent is printed on each plot — here $\nu = 0.593$, the value expected for a flexible coil in a good solvent.

Procedure. Include the samples you want, pick the focused sample/fraction with the Sample and Fraction selectors, choose the R_g / R_h / M_w / A_2 source per row, read ρ and its Shape label in the table, and switch to the Scaling sub-tab for the log–log plots and fitted exponents. See the Theory-and-Equations-Guide for the equations and theory.

11. Exporting Results

Every analysis result can be written to a CSV that OriginLab imports directly. Each file carries the four Origin label rows — **Long Name**, **Units**, **Comments**, **Parameters** — so axis labels and units populate on import. Tabular outputs (correlogram fits, distributions, Rayleigh ratios) are written column by column with units; scalar results (M_w , R_g , Γ , ...) as single-value labeled columns. Zimm datasets use a wide layout (one $Kc/\Delta R\theta$ column per concentration, the concentration in the Parameters row). Missing values are empty cells. An uncalibrated result writes “uncalibrated, arbitrary scale” into the **Comments** cell of the affected columns only — never changing the row count, so the Origin import stays intact.

From the interface. Each analysis view has an **Export CSV...** button that writes the result currently on screen: the DLS **Correlogram**, **Distribution**, Γ vs q^2 , **D** vs c , and **Depolarized (DDLS)** sub-tabs (the DDLS file carries the per-angle FVV/FVH and Dr , the combined $Dt/Dr/Rh, t/\tau_{rot}$, and — when available — the rod and sphere shape dimensions, each flagged “MODEL ... not measured” in its Comments cell); the SLS **Analysis** panel (the button exports the last method you ran — Zimm/Berry, Debye, Guinier, single-angle, calibration-free A_2 , or the

excess Rayleigh series); and the Cross-Sample **Scaling** tab (the R_g/R_h - M_w and A_2 - M_w points and fits). The button is enabled once a result has been computed. When a view holds several curves at once (an NNLS+CONTIN distribution overlay, or both size and A_2 scaling plots), the export writes one file per curve, suffixed by method/quantity.

12. Visualization

Every analysis can be plotted from the same result object it returns, so what is plotted is exactly what was computed: correlogram fits with residuals, size distributions, Γ -vs- q^2 diagnostics, Debye plots, and the full Zimm/Berry grid. Distributions from different methods can be overlaid (e.g. NNLS and CONTIN on one axis), making their trade-off visible at a glance.

Data-quality and reliability **flags are interface overlays**, not part of the figure: a saved plot image is always clean. Where a result is apparent or uncalibrated, the interface marks it next to the plot, so a number can't be read off a chart without its caveat.

Appearance (Settings tab). The **Theme** is *System* (follows your OS light/dark setting) by default; choosing *Light* or *Dark* explicitly **overrides** the OS so the application stays in that mode regardless of system preferences. The **Plot palette** (Okabe-Ito colorblind-safe / tab10 / grayscale) applies to plots drawn afterwards. Within the analysis tabs, the **stacked plots and their control columns are resizable** — drag the **splitter grips** (the visible three-dot handles) between them, including the fit/residual divider, which carries a persistent grip line. The workspace panel on the left is **resizable** (drag the divider) and long sample names show in full on hover; the window itself can be freely resized (tab contents scroll when the window is small).